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Morehouse, III et al.

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(54) **FILTER ELEMENT LOCKING MECHANISM**

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See application file for complete search history.

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B01D 35/30 (2006.01)

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(2013.01)

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CPC B01D 29/15; B01D 35/306; B01D

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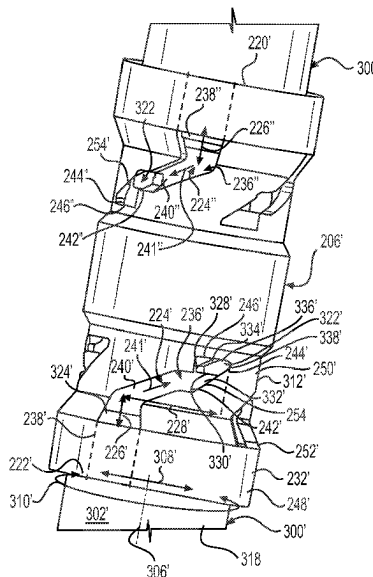
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ABSTRACT

A method of assembling a canister filter system includes inserting a first filter component into a second filter component, and continuing the insertion until a first tab of either the first filter component or the second filter component contacts the other of the first filter component and the second filter component.

12 Claims, 10 Drawing Sheets



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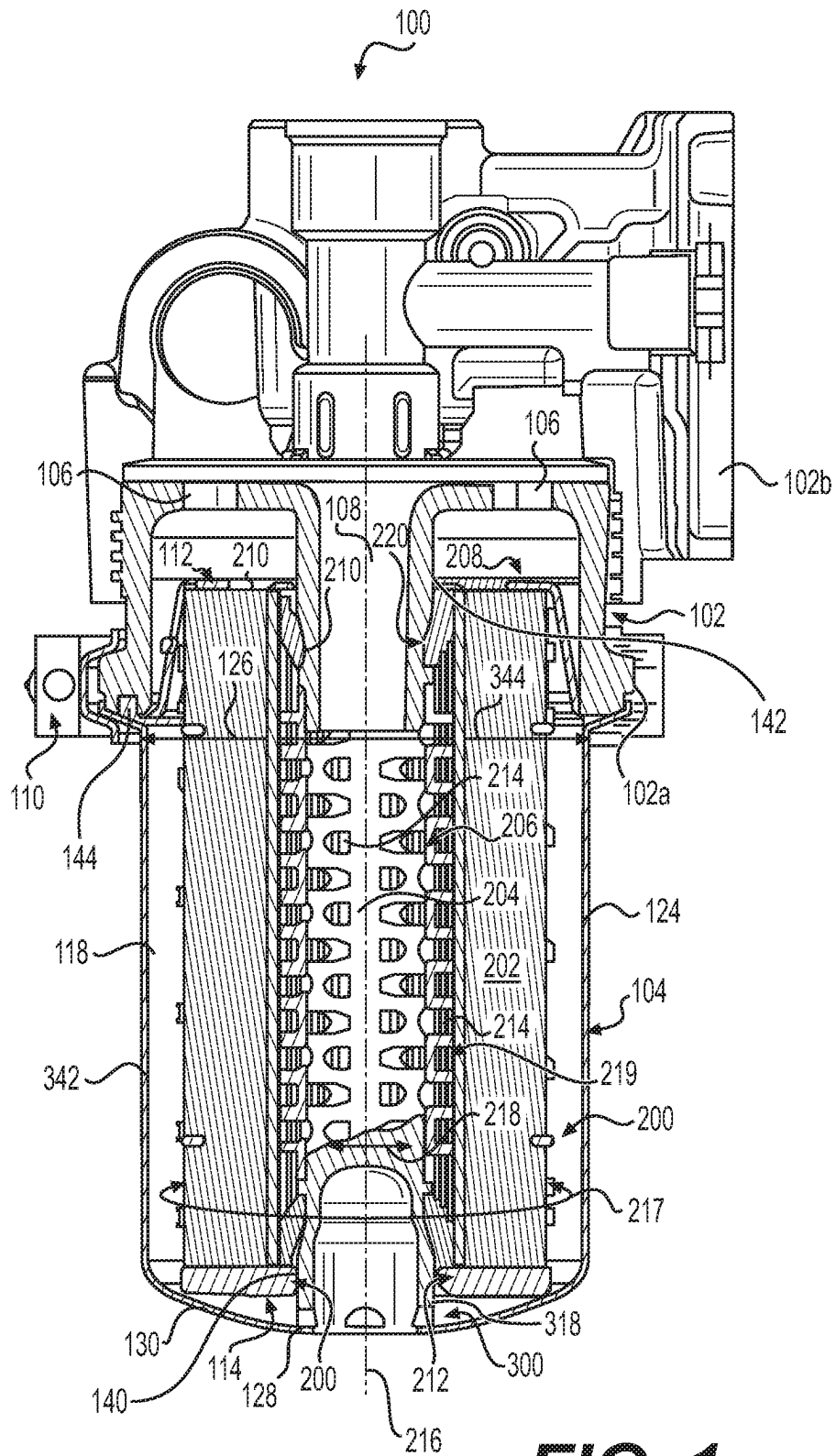
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**FIG. 1**

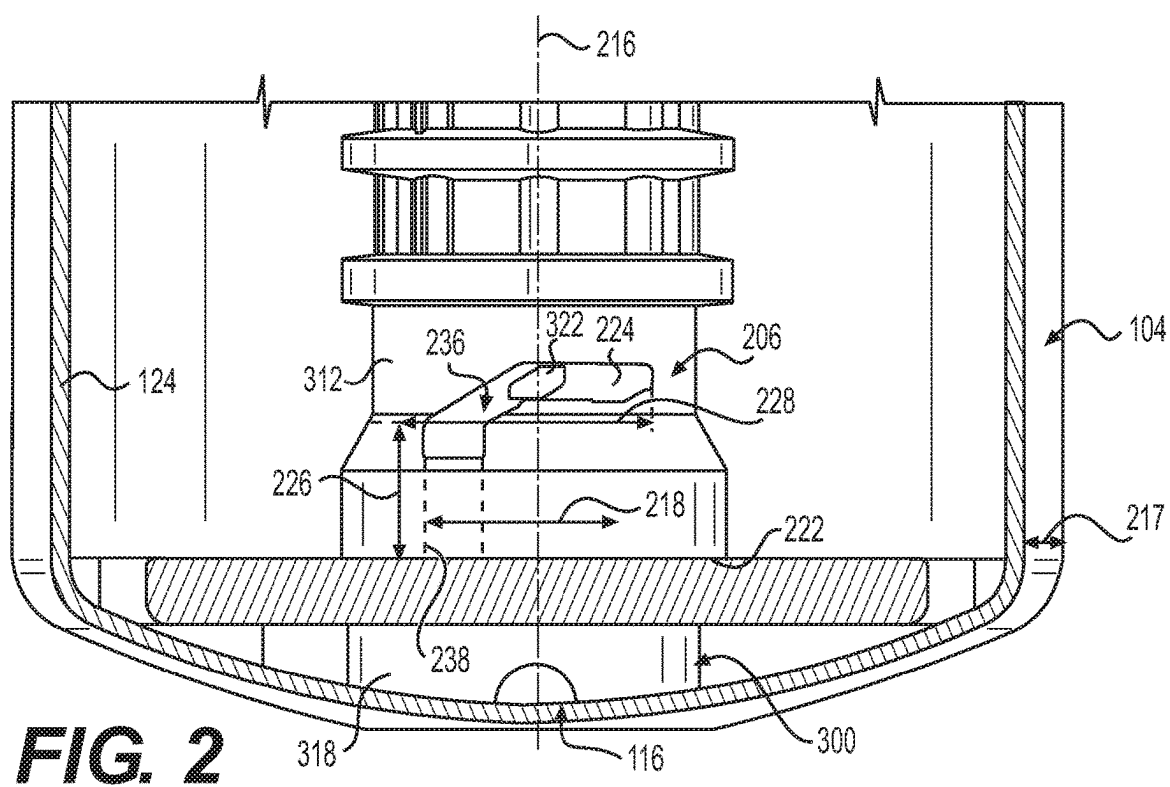


FIG. 2

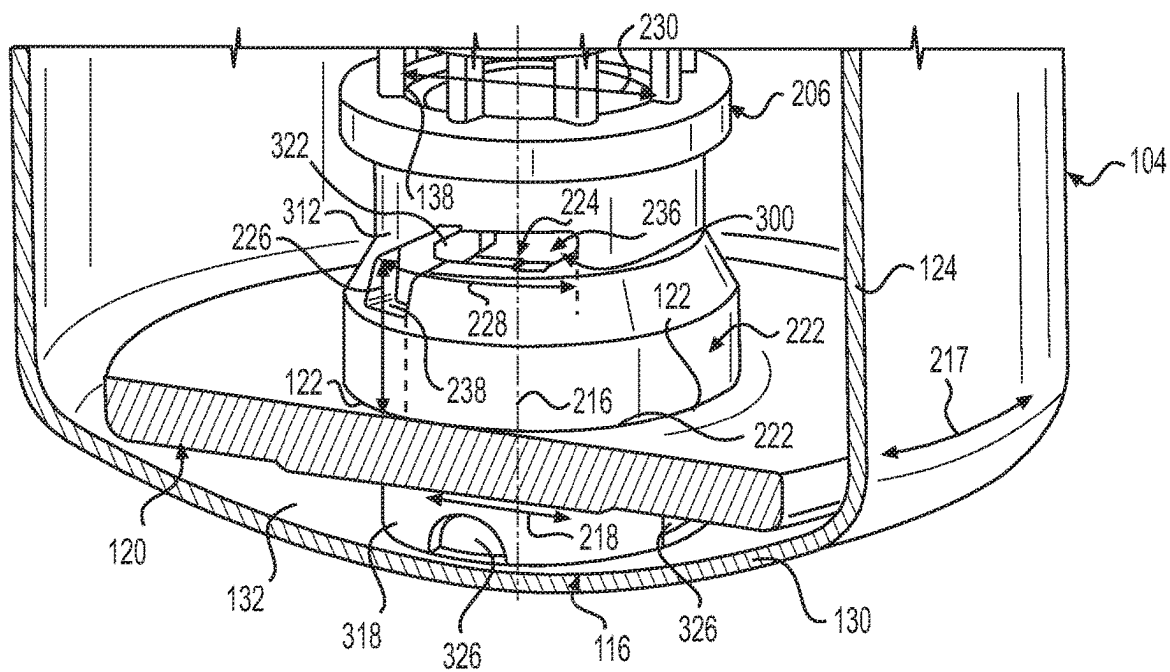


FIG. 3

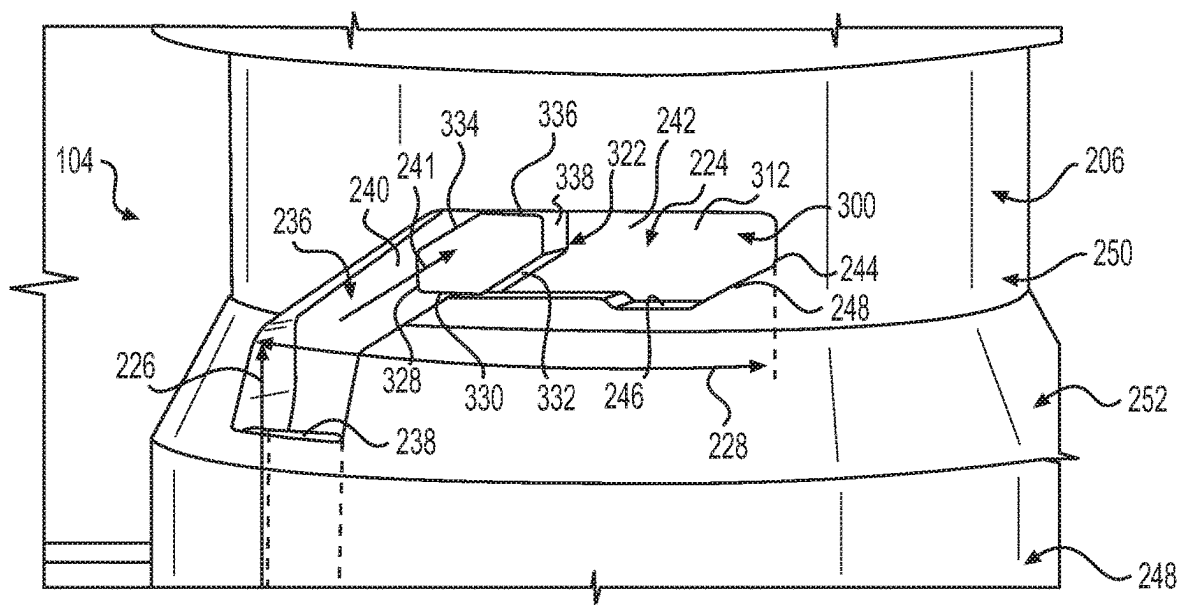


FIG. 4

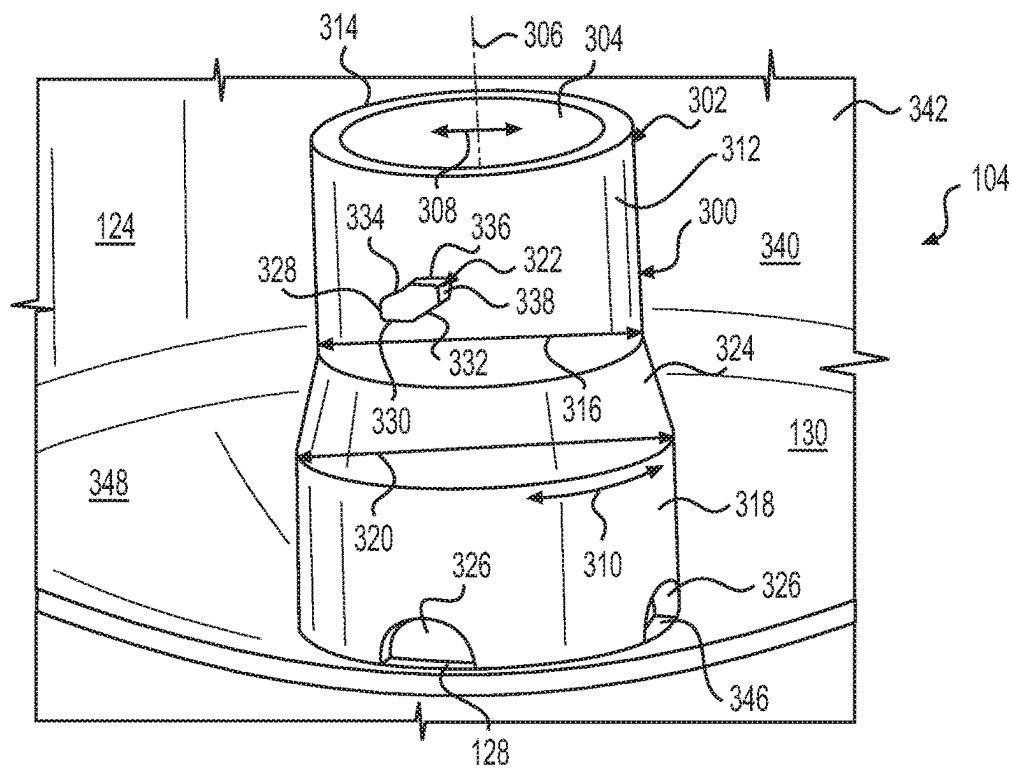
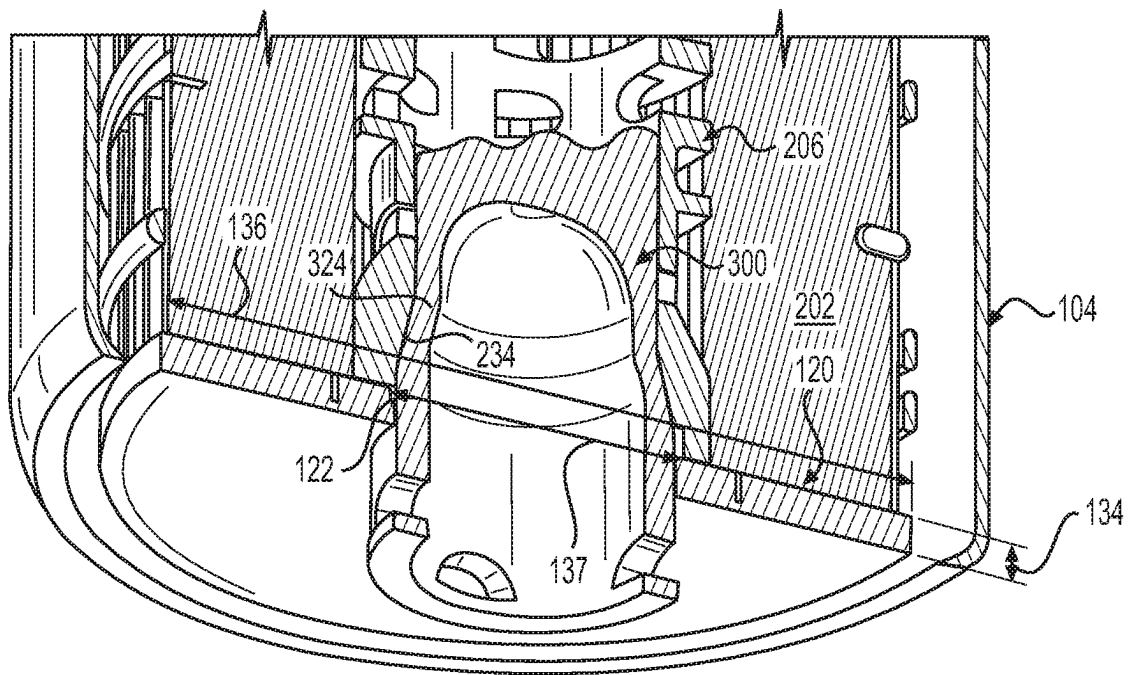
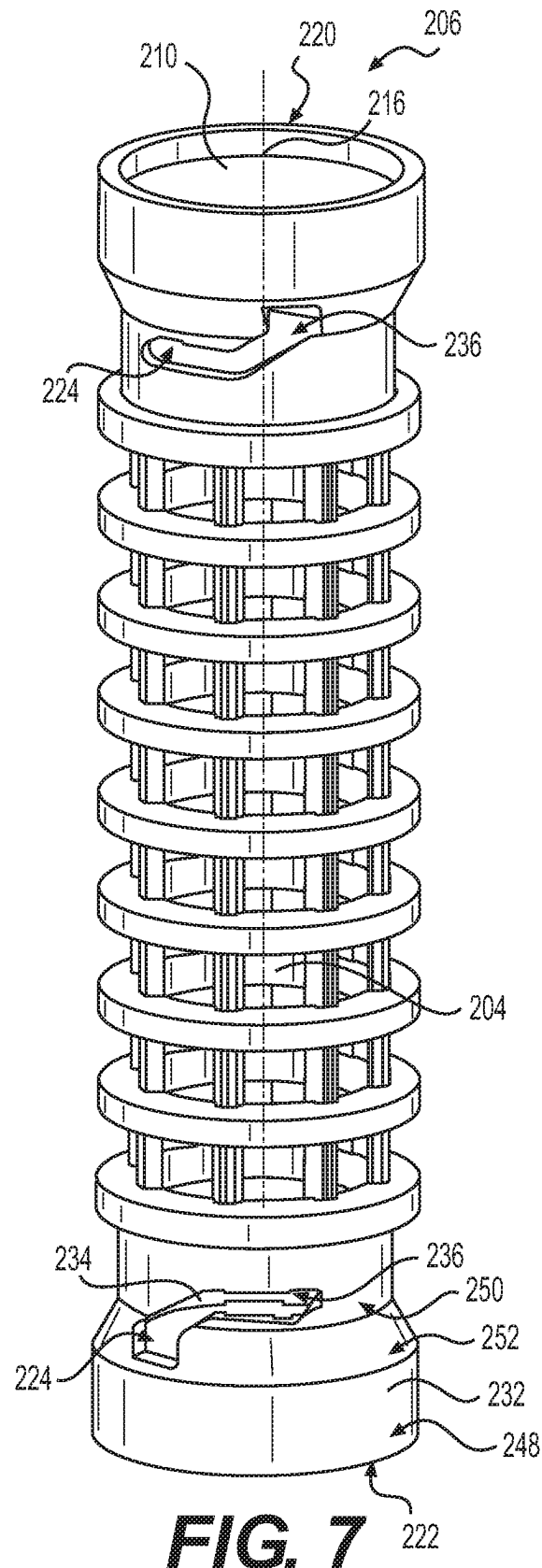


FIG. 5

**FIG. 6**



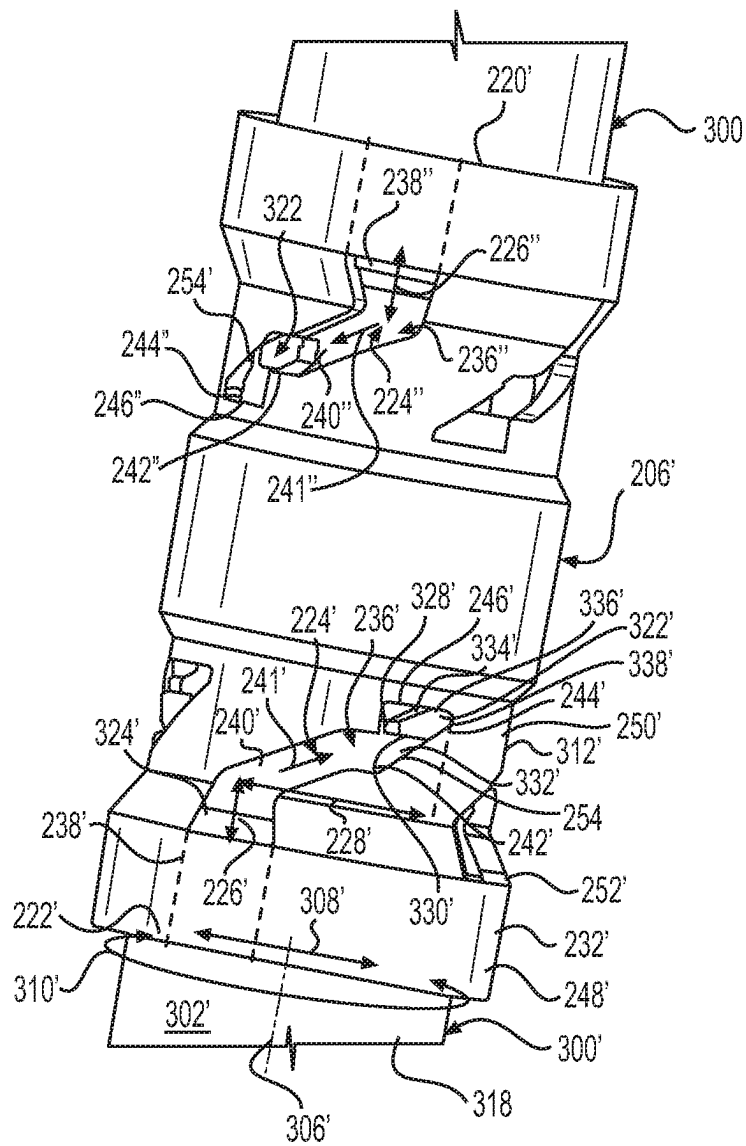
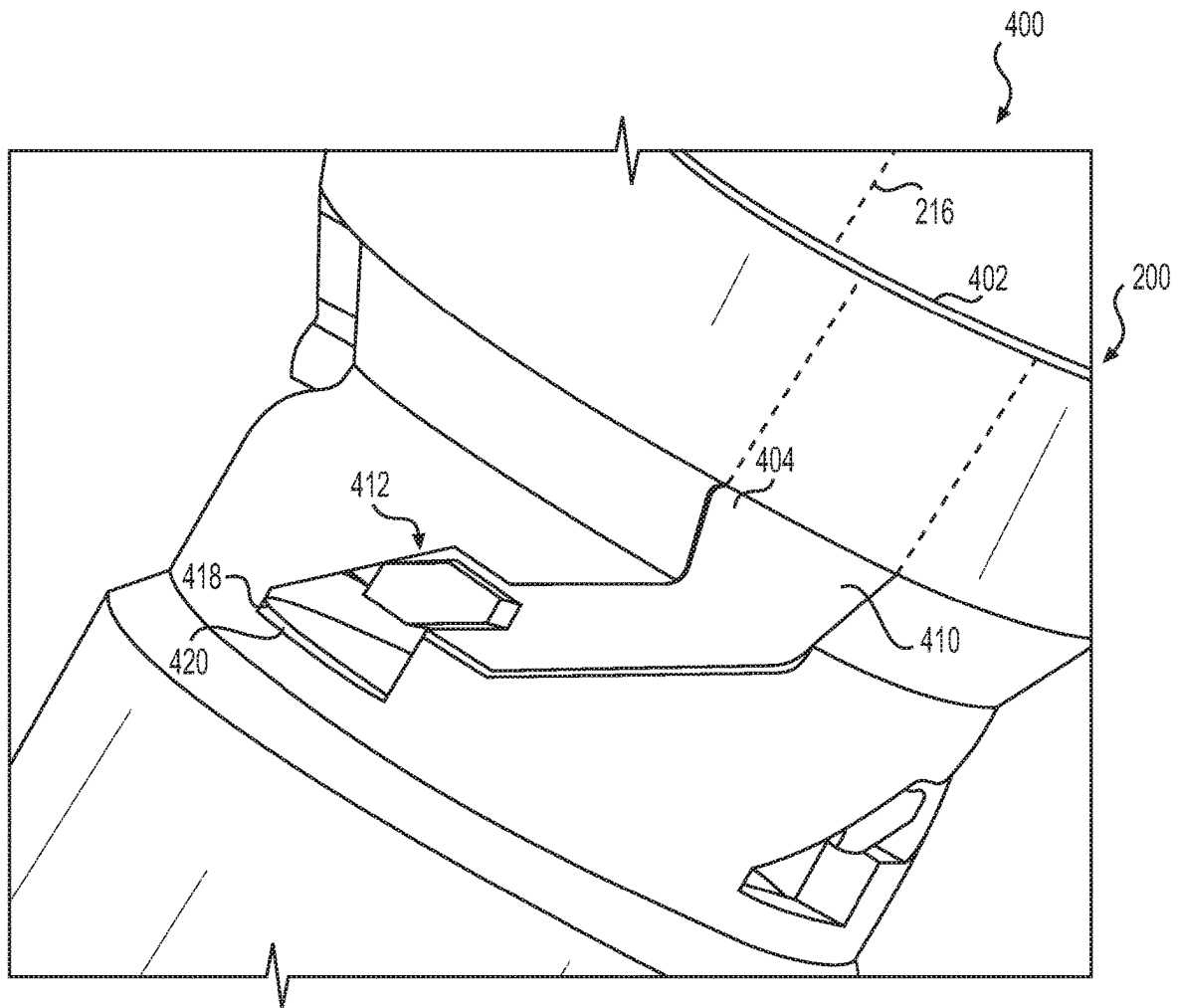
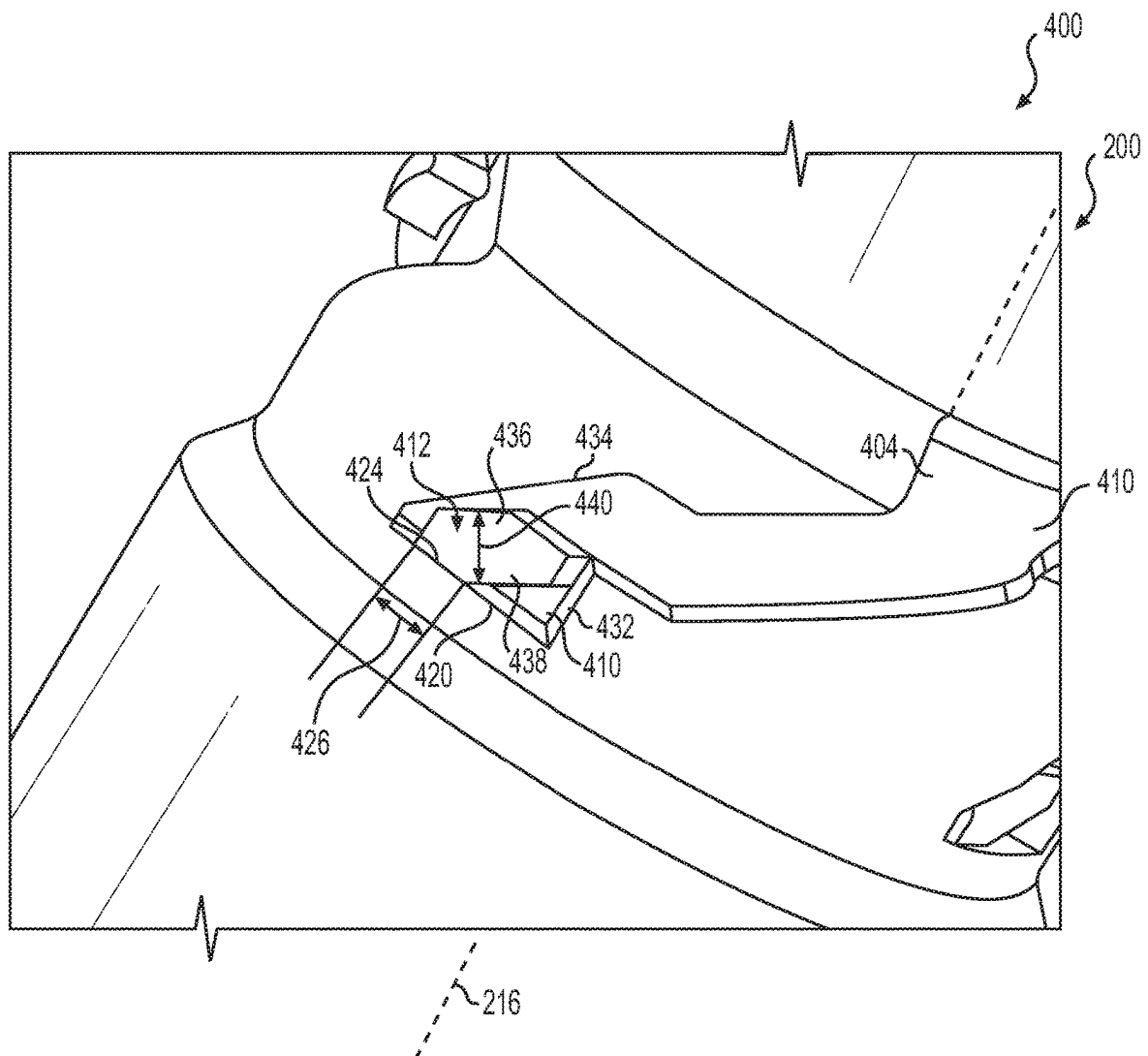
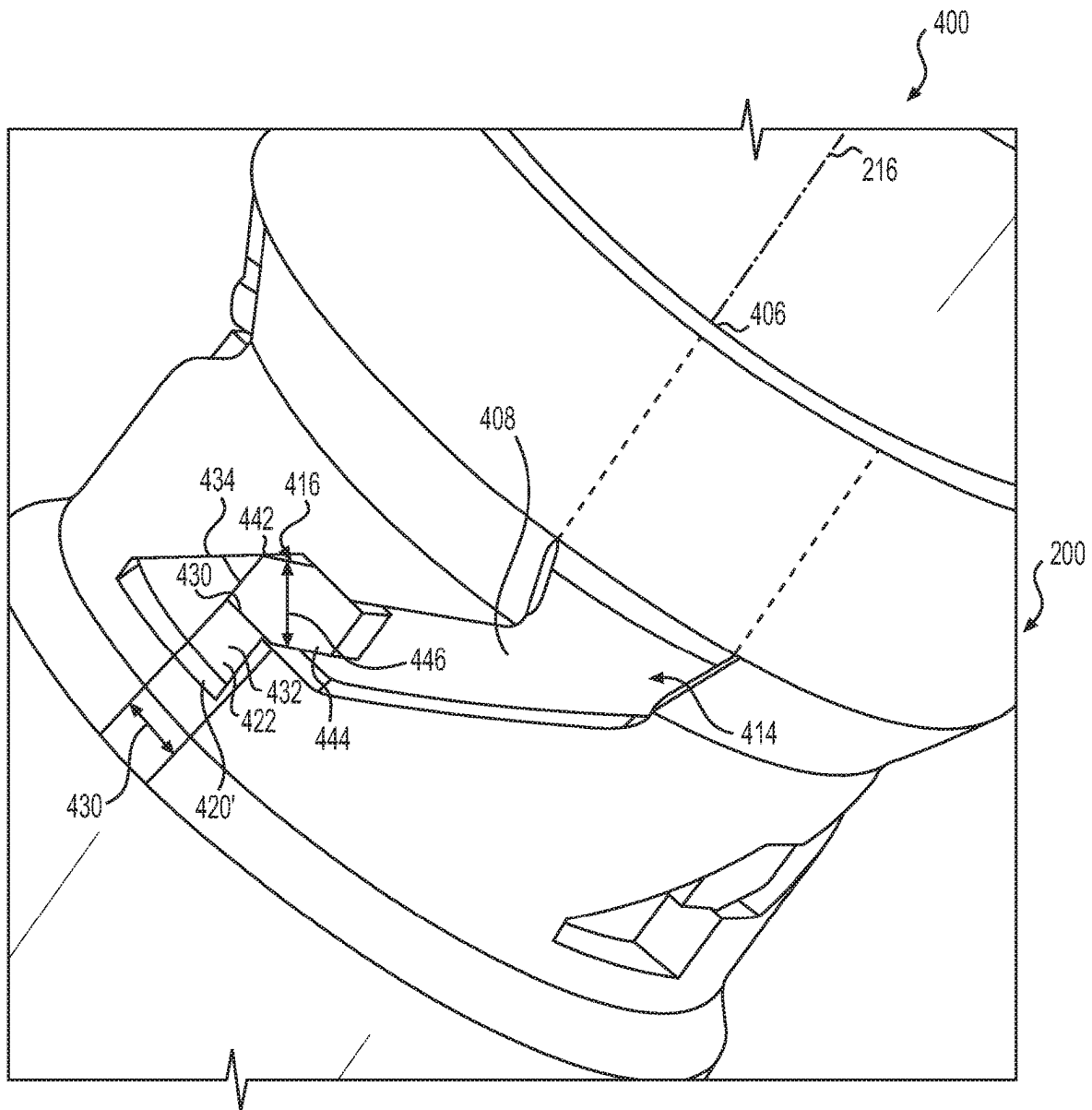
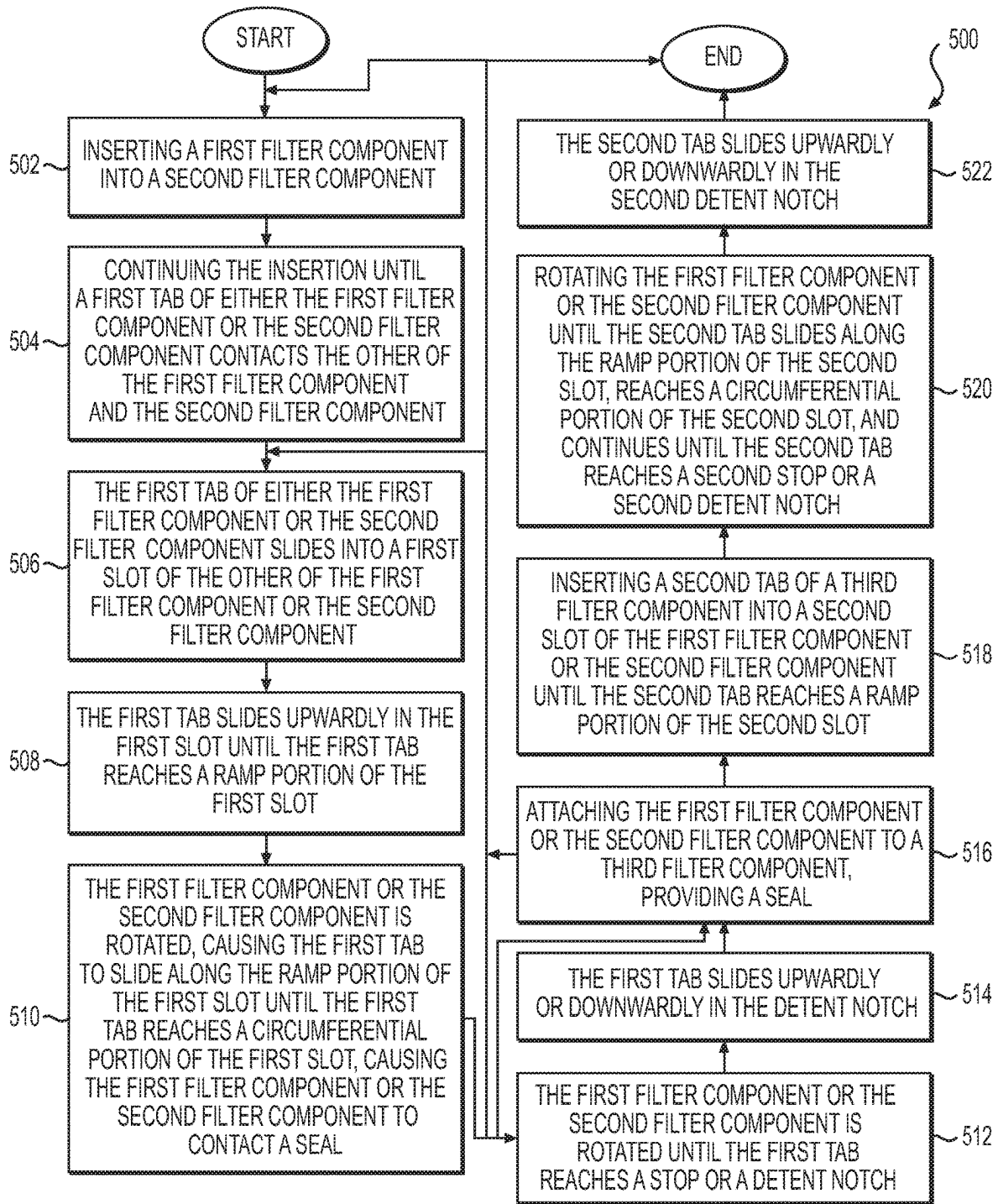


FIG. 8

**FIG. 9**

**FIG. 10**

**FIG. 11**

**FIG. 12**

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FILTER ELEMENT LOCKING MECHANISM**CROSS-REFERENCE TO RELATED APPLICATIONS**

The present disclosure is a divisional of U.S. patent application Ser. No. 16/799,356, filed Feb. 24, 2020, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates generally to canister style filter systems that employ a replaceable filter element. More specifically, the present disclosure relates to a filter element that includes a locking feature for retaining the filter element in a desired position while providing a seal that helps to ensure that dirty fluid is filtered by the filtering medium of the filter element.

BACKGROUND

Liquid filter systems are known for filter various fluids such as gas, oil, diesel fuel, etc. to remove contaminants from these fluids. In diesel engines, for example, a fuel line filter is used to separate out water and debris from the fuel. These contaminants may accumulate in a lower portion of the filter housing (may also be referred to as a canister). Typically, the center tube of the filter element provides support for the filter media but may not always provide the desired location and retention.

U.S. Pat. Application Publication No. 20060207948 to Hacker et al. discloses a fluid filter assembly including a housing, a service cover, a center tube removably secured to the service cover, and a filter cartridge removably sealed and circumscribing the center tube, and a seal arrangement. The seal arrangement is between the center tube and portions of the housing to close a drainage change to the flow of a clean fluid flow therethrough, when the fluid filter assembly is operating to filter the fluid.

During normal operation in Hacker, the fluid filter assembly operates to allow fluid to flow into the housing through an inlet channel, through the filter cartridge, through openings in the center tube, and out of the housing through the outlet channel. Methods for servicing include removing a service cover from a housing to remove, together with the service cover, a center tube, and open a drainage flow passageway from the housing. Next a filter cartridge is removed from the center tube, and a new filter cartridge is operably mounted on the center tube. Next, the service cover with the center tube having the new filter cartridge is operably mounted in the housing to close the drainage flow passageway.

In Hacker, methods of filtering will direct fluid to be filtered into a housing having a removable and replaceable filter cartridge; then direct the fluid through a tubular region of filter media in the cartridge; then through fluid openings in a center tube; and into a clean fluid flow passageway. Example methods include preventing fluid to bypass the filter media by removably sealing the filter cartridge to the center tube. Example methods will also include preventing fluid from flowing into a drainage passageway by removably sealing the center tube to other portions of the filter housing. Systems utilizing filter assemblies described in Hacker include fuel systems, lube systems, and hydraulic systems.

Hacker fails to disclose a feature on the center tube that may be used to both position and retain the filter element in

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a desired position while also providing a seal that forces dirty fluid to pass through the filter medium of the filter element before exiting the filter element.

SUMMARY OF THE DISCLOSURE

An interface for use with a filter system according to an embodiment of the present disclosure may comprise a filter element that includes at least partially a cylindrical configuration and that defines a longitudinal axis, and a radial direction. The filter element may comprise an annular filter media defining a central passage, a center tube that is disposed in the central passage of the annular filter media that defines a central reservoir, and the annular filter media surrounds the center tube and the central reservoir. A first open end may be joined to the center tube disposed along the longitudinal axis, the first open end also defining a first locking slot, and a second open end may be joined to the center tube opposite the first open end disposed along the longitudinal axis. The second open end may also define a second locking slot. A first pedestal may be provided that includes a first tab that is disposed in the first locking slot, and a second pedestal may also be provided that includes a second tab that is disposed in the second locking slot. The second tab may be differently configured as the first tab.

A filter seal for use with a canister filter system according to an embodiment of the present disclosure may comprise an annular body defining an outer diameter, a thickness, and a central hole.

A method of assembling a canister filter system according to an embodiment of the present disclosure may comprise inserting a first filter component into a second filter component, and continuing the insertion until a first tab of either the first filter component or the second filter component contacts the other of the first filter component and the second filter component.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a front sectional view of a filter assembly that includes a filter base, a canister, and a filter element including a center tube with a locking feature according to an embodiment of the present disclosure, and a pedestal that is configured to mate with the locking feature.

FIG. 2 is a front sectional view of the bottom portion of a filter assembly similar to that of FIG. 1 with the annular filter media removed, showing the mating of the pedestal with the locking feature of the center tube according to an embodiment of the present disclosure.

FIG. 3 is a perspective view of the filter assembly of FIG. 2 illustrating how the pedestal rests or extends from the bottom of the canister.

FIG. 4 is an enlarged perspective view of the mating of the tab of the pedestal with the locking feature of the center tube.

FIG. 5 shows the canister and pedestal shown in isolation from the filter assembly of FIG. 3.

FIG. 6 is a bottom oriented perspective view of the filter seal showing its aperture that is configured to mate with the cylindrical or conical surface of the pedestal.

FIG. 7 is a perspective view of the center tube shown in isolation from the filter assembly of FIG. 1.

FIG. 8 is a perspective view showing conceptually how a center tube such as the one similar in construction to the center tube of FIG. 7 may be configured to mate with the base at the top end and the pedestal at the bottom end of a filter assembly.

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FIG. 9 is an enlarged detail view showing the mating of the locking slot of the center tube and pedestal similar to the bottom portion of FIG. 8 except the pedestal is shown near the top in FIG. 9. The tab of the pedestal is shown being rotated during an assembly or locking operation.

FIG. 10 illustrates that the tab of the pedestal of FIG. 9 is small enough to fall into the detent notch.

FIG. 11 is an enlarged detailed view showing the mating of the locking slot of the center tube and pedestal similar to the top portion of FIG. 8. The tab of the pedestal is shown to be too large to fit into the detent notch.

FIG. 12 is a flow chart containing a method of operation or assembly associated with FIGS. 9 thru 10.

DETAILED DESCRIPTION

Reference will now be made in detail to embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. In some cases, a reference number will be indicated in this specification and the drawings will show the reference number followed by a letter for example, 100a, 100b or a prime indicator such as 100', 100" etc. It is to be understood that the use of letters or primes immediately after a reference number indicates that these features are similarly shaped and have similar function as is often the case when geometry is mirrored about a plane of symmetry. For ease of explanation in this specification, letters or primes will often not be included herein but may be shown in the drawings to indicate duplications of features discussed within this written specification.

First, a filter system will now be described to give the reader the proper context for understanding how various embodiments of the present disclosure are used. It is to be understood that this description is given as exemplary and not in any limiting sense. Any embodiment of an apparatus or method described herein may be used in conjunction with any filter system.

Then, a filter element that may include a center tube with a locking feature according to various embodiments will be discussed. This feature may be located in the bottom end of a liquid filter assembly with a reusable housing (may be referred to as a canister), and it may position the filter element radially and axially in the canister (may also be referred to as the housing) while also separating the clean fluid from the dirty fluid on different sides of the filter medium via a seal. In fuel-water separators, the positioning geometry may be configured to create a chamber for collecting water (i.e. a water bowl) or debris by preventing the element from extending to the bottom of the canister.

FIG. 1 illustrates a canister filter system 100 that may use a filter element 200, and a pedestal 300 according to various embodiments of the present disclosure.

The canister filter system 100 may include having a base 102 (shown to be split up into a base adapter 102a and a filter base 102b, but could be one integral component), a canister 104, a pedestal 300, and a filter element 200. The canister filter system 100 may be used to filter fluids such as diesel or gasoline or other liquid fuels, lubrication oil, hydraulic fluid for hydraulic power systems, transmission fluid, or even possibly intake air for an engine. The canister filter system 100 may also be used as a fuel/water separator filter. The canister filter system 100 with the features described herein could be adapted by those of ordinary skill in this art to serve many different purposes and suit many other applications.

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The base 102 includes an inlet channel 106 for fluid to enter into the canister filter system 100, and an outlet channel 108 for fluid to exit from the canister filter system 100. A clip 110 is provided to attach the canister 104 to the base 102. Other attachment structure such as threads may be used.

The canister 104 includes a top open end 112 and a bottom open end 114 as shown in FIG. 1 or a bottom closed end 116 as shown in FIGS. 2 and 3.

The filter element 200 may take many different forms to suit a particular application. In the illustrated embodiment, the filter element 200 is well suited for filtering fuel or lubrication oil. The filter element 200 may include annular filter media 202 circumferentially surrounding a central reservoir 204 defined by a center tube 206. Axial ends of annular filter media 202 are shown to be sealed by end caps.

A top end cap 208 may define an axial open end of filter element 200. The top end cap 208 is termed "open" because it includes an opening 210 for allowing passage of fluid.

On the other hand, the bottom end cap 212 defines an axial closed end of filter element 200. The bottom end cap 212 is termed "closed" because it prevents any fluid outside the filter element 200 adjacent the axial end of the annular filter media 202 from flowing unfiltered into center tube 206.

The top end cap 208 and the bottom end cap 212 may each be joined to the center tube 206 via welding, adhesives, etc. Alternatively, several or all of center tube 206, the top end cap 208, and the bottom end cap 212 may be constructed as unitary components. Conversely, the bottom end cap 212 and/or the top end cap 208 may be separate components from the center tube 206, etc. Further details of the closed configuration of the bottom of the canister filter system 100 and the filter element 200 will be discussed later herein.

In operation, fluid to be filtered enters from the inlet channel 106 and flows to the annular cavity 118 between canister 104 and the annular filter media 202. The fluid then passes into and through filter media 202, then into the center tube 206 through the perforations 214 shown therein in FIG. 1.

Then, the fluid exits center tube 206 through the top end cap 208 and opening 210 into the outlet channel 108. The sealed construction at the bottom of the filter element 200 helps to define the fluid channels into and out of the annular filter media 202, preventing any fluid from flowing directly to outlet channel 108 and bypassing the annular filter media 202. To that end, sealing features may be provided that will be discussed in detail later herein. Moreover, it may be desirable to create a chamber (e.g. a water bowl in fuel-water separators, a drain reservoir, etc.) between the bottom of the filter element and the bottom of the canister. So, a positioning feature may be provided as will be discussed later herein.

Referring now to FIGS. 1 thru 3, a canister filter system 100 according to various embodiments of the present disclosure that provides locking and/or positioning feature(s) will now be discussed.

The canister filter system 100 may comprise a filter element 200 that includes at least partially a cylindrical configuration and that defines a longitudinal axis 216, a circumferential direction 217, and a radial direction 218. The filter element 200 may comprise an annular filter media 202 defining a central passage 219 and a center tube 206 that is disposed in the central passage 219 of the annular filter media 220 that defines a central reservoir 204. Thus, the annular filter media 202 surrounds the center tube 206, and the central reservoir 204.

As best seen in FIG. 1, the filter element 200 may further include a top open end 220 joined to the center tube 206

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disposed along the longitudinal axis **216**. The top open end **220** includes an opening **210** that allows fluid to flow from the central reservoir **204** to the outside of the filter element **200**.

Similarly, the filter element **200** may include a bottom open end **222** joined to the center tube **206** opposite the top open end **220** that is also disposed along the longitudinal axis **216**. Thus the bottom open end **222** allows insertion of the pedestal **300**.

The canister filter system **100** may also include a canister **104** that includes a top open end **112** (see FIG. 1), and a bottom closed end **116** (see FIGS. 2 and 3) relative to the longitudinal axis **216**, and a pedestal **300** that rests on the bottom closed end **114** of the canister **104**. This may not be the case in other embodiments of the present disclosure such as in FIG. 1 wherein the pedestal **300** is molded integral with the canister **104**.

Looking now at FIGS. 2 thru 5, the pedestal **300** may include an at least partially annular body **302** (may be completely annular with a thru-hole **304** extending from end to end as best seen in FIG. 5 or not such as shown in FIG. 1) defining a longitudinal axis **306**, a radial direction **308**, and a circumferential direction **310** (see FIG. 5, may be concentric with the filter element once assembled).

Focusing on FIG. 5, the body **302** may include a top annular portion **312** terminating at a top free end **314** (regardless of whether it is open or closed) and may also define a top diameter **316** (i.e. top outer diameter), and a bottom annular portion **318** defining a bottom diameter **320** (i.e. bottom outer diameter) that is greater than the top diameter **316**.

A tab **322** may extend radially from the top annular portion **312**, and a filter seal **120** may be provided that defines an aperture **122** that is configured to mate with the bottom annular portion **318** of the pedestal **300** (see also FIG. 6).

With continued reference to FIG. 5, the pedestal **300** may rest on the bottom closed end **114** of the canister **104**. The filter seal **120** may be disposed below the tab **322** of the pedestal **300** and around the bottom annular portion **318** while contacting the canister **104**.

More specifically, the canister **104** comprises an outer annular wall **124** that defines an outer diameter **126** (see FIG. 1) that is greater than the bottom diameter **320** of the pedestal **300**, a support wall **128** that is configured to support the bottom annular portion **318** of the pedestal **300**, and an arcuate wall **130** (other configurations are possible) that connects the support wall **128** to the outer annular wall **124** (see FIG. 5).

As shown in FIG. 3, the filter seal **120** may contact the arcuate wall **130**, forming a drain reservoir **132**. More particularly, the bottom open end **222** of the center tube **206** may impinge upon the filter seal **120**, creating a fluid tight seal between center tube **206** and the filter seal **120**, and another fluid tight seal between the filter seal **120** and the arcuate wall **130** of the canister **104**.

Looking at FIG. 6, a ratio of the outer diameter **136** to the thickness **134** of the filter seal **120** of the filter seal may range from 10.0 to 30.0, while a ratio of the outer diameter **136** to the hole diameter **137** may range from 1.5 to 3.0. This geometry in addition to the elastomeric composition of the filter seal may provide the suitable balance between rigidity and flexibility to allow the center tube and pedestal to be connected while providing the water tight seals. For example, a material of urethane having a durometer of 55 to 65 Shore A (e.g. 60 Shore A) may be employed.

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Looking at FIGS. 2 thru 4, the center tube **206** includes a locking slot **224** that extends axially from the bottom open end **222** of the center tube **206** an axial distance **226**, and then extends circumferentially a circumferential distance **228**.

Referring to FIGS. 3 and 5, the center tube **206** may define an inner diameter **230** that is slightly greater (i.e. 0.015 of an inch to 0.030 an inch of clearance) than the top diameter **316** of the pedestal **300** that is disposed within the inner diameter **230**, and the tab **322** of the pedestal **300** may be disposed in the locking slot **226**, and may be configured to guide the movement of the center tube **206**. This arrangement may provide a positioning/centering function. Though not shown, an O-ring or other seal may be provided at this interface **138** to prevent fluid from bypassing the annular filter media **202** in some embodiments such as when the annular filter media does not extend axially past the locking slot.

As best seen in FIG. 5, the pedestal **300** may include a flared annular portion **324** (e.g. a conical shape, an arcuate shape, etc.) that connects the top annular portion **312** to the bottom annular portion **318**. This feature may be omitted in other embodiments. As best seen in FIG. 6, the radially inner surface **234** of the center tube **206** may have a matching shaped (e.g. angled or tapered) to ease installation of the center tube/filter element onto the pedestal. This may provide guidance as the tabs enter the locking slots.

Referring once more to FIG. 5, the bottom annular portion **318** may be attached to the canister **104**. For example, when the canister and the pedestal are formed by thermoplastic injection molding (e.g. using a polyurethane material, nylon material, etc.), the pedestal may be ultrasonically welded to the canister, molded integrally with the canister, snapped onto the canister, threaded onto the canister etc. Also, a protuberance may extend from the canister about which the pedestal is centered, etc. Other manufacturing methods are possible including sheet metal fabrication, etc. When metal is employed, the pedestal may be attached to the canister via welding, brazing, threading, etc.

The bottom annular portion **318** may define at least one thru-slot **326** that extends radially and possibly axially downwardly through the bottom annular portion **318**. This may allow water or debris to exit from the central reservoir **204** of the center tube **206** to the drain reservoir **132** (e.g. see FIG. 3).

Now, a filter element **200** that may be used with the that includes at least a partially annular configuration (e.g. conical, cylindrical, other bodies of revolution, etc.) and that may be used with the canister filter system **100** just described will be discussed with reference to FIGS. 1 thru 4, and 7.

Starting with FIGS. 1 and 7, the filter element **200** may define a longitudinal axis **216**, a radial direction **218**, and a circumferential direction **217**. Moreover, the filter element **200** may comprise an annular filter media **202** defining a central passage **219**.

A center tube **206** may be disposed in the central passage **219** of the annular filter media **202** that defines a central reservoir **204**. Accordingly, the annular filter media **202** surrounds the center tube **206** and the central reservoir **204**. A top open end **220** may be joined to the center tube **206** that is disposed along the longitudinal axis **216**. The top open end **220** includes an opening **210** allowing fluid to flow from the central reservoir **204** to the outside of the filter element **200** or vice versa.

Similarly, a bottom open end **222** may be joined to the center tube **206** axially opposite the top open end **220**

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disposed along the longitudinal axis **216**. As best seen in FIG. **3**, the bottom open end **222** may define a radially outer surface **232**, and a radially inner surface **234** that is in communication with the central reservoir **204**.

Looking at FIGS. **2** thru **4**, a locking feature **236** may be disposed proximate to the bottom open end **222** of the center tube **206**. The locking feature **236** may include an entrance slot **238** that is disposed on the radially inner surface **234** of the center tube **206** that extends axially upwardly from the bottom open end **222**. This entrance slot is configured to allow the center tube **206** to slide over the tab **322** of the pedestal **300** alluded to earlier herein during assembly.

As best seen in FIG. **4**, the locking feature **236** may further comprise a ramp slot **240** that extends axially upwardly and circumferentially counterclockwise from the entrance slot **238** along a predetermined direction **241**. The ramp slot **240** is in communication with the entrance slot **238** so that as the tab **322** of the pedestal **300** moves upwardly in the entrance slot **238** when the center tube **206** moves downwardly, the tab **322** will eventually arrive at the ramp slot **240**. Twisting the center tube **206** circumferentially will cause the center tube **206** to move downwardly, impinging upon the filter seal **120** (e.g. see FIG. **3**). In other embodiments, this arrangement may be reversed such that the ramp slot extends axially upwardly and circumferentially clockwise. In such a case, the center tube will need to be twisted in the opposite direction.

With continued reference to FIG. **4**, the locking feature **236** may further comprise a circumferential locking slot **242** that extends circumferentially counterclockwise from the ramp slot **240**. The circumferential locking slot **242** is in communication with the ramp slot **240**, and terminates at a stop surface **244**. Consequently, as the center tube **206** is twisted circumferentially, the axial position of the center tube is substantially fixed by the tab **322** of the pedestal **300**. Again, this arrangement may be reversed such that the circumferential locking slot extends in the clockwise direction, necessitating that the center tube be rotated in the opposite direction.

Continued twisting of the center tube **206** in the counterclockwise direction will allow the tab **322** to contact or nearly contact the stop surface **244**, at which time the upward force exerted by the filter seal **120** causes the center tube **206** to move slightly upwardly until the tab **322** engages a detent notch **246** that extends axially downwardly from the circumferential locking slot **242** being in communication therewith. Now, the center tube **206** is locked into position both axially, radially and circumferentially against unintentional movement.

An angled surface **248** may extend from the detent notch **246** to the stop surface **244**, matching the shape of the tab **322**. Hence, the angled surface **248** may parallel to the predetermined direction **241** that the ramp slot **240** extends. Likewise, the stop surface **244** may extend axially to match shape of the tab **322**.

Other configurations are possible for these various features.

Looking at FIGS. **4** and **7**, the center tube **206** has a necked down configuration at the bottom open end **222** including an enlarged bottom annular portion **248**, a reduced top annular portion **250**, and a transitional annular portion **252** therebetween. The entrance slot **238** is formed by the enlarged bottom annular portion **248**, and the transitional annular portion **252**. The ramp slot **240** is formed by the transitional annular portion **252**, and the reduced top annular

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portion **250**. The circumferential locking slot **242**, and the detent notch **246** are formed by the reduced top annular portion **250**.

Again, other configurations for these features are possible in other embodiments of the present disclosure.

Looking at FIG. **7**, it can be seen that the center tube **206** (and therefore the filter element **200**), are constructed so that both ends are similarly or identically constructed. So, the center tube **206** may be rotated about an axis that extends through axial midpoint of the center tube and perpendicular to the longitudinal axis **216** an amount of 180 degrees and still be able to be installed with the pedestal **300**. This may not be the case in other embodiments of the present disclosure. In such a case, the top locking feature may or may not be used to attach the base to the center tube. An example, of the use of both top and bottom locking features in a canister filter system will be discussed later herein.

The top portion of the entrance slot **238**, and the entirety of the ramp slot **240**, the circumferential locking slot **242**, and detent notch **246** extend completely radially through the center tube **206**. On the other hand, the lower portion of the entrance slot **238** does not extend completely radially through the center tube **206** (denoted by dotted lines). Thus, a side action may form the thru-portions of these features while a bottom core may form the blind portion of the entrance slot and contacts the side action. So, the center tube may be manufactured using a thermoplastic injection molding process. A nylon, a polyurethane, or any other suitable material may be used to form the center tube.

Other configurations for these features may be employed to facilitate the use of other manufacturing processes, etc.

Next, a pedestal **300** that may be used with a canister filter system **100**, and a filter element **200** for positioning and retaining the filter element **200** in the canister filter system **100** will now be discussed with reference to FIGS. **4** and **5**.

The pedestal **300** may comprise an at least partially annular body **302** defining a longitudinal axis **306**, a radial direction **308**, and a circumferential direction **310**.

The body **302** may include a top annular portion **312** terminating at a top free end **314** as earlier described herein. A tab **322** extends radially from the top annular portion, **312**. The tab **322** includes a first axial surface **328**, a bottom circumferential surface **330**, and a bottom ramp surface **332** extending from the bottom circumferential surface **330**. The first axial surface **328** is configured to engage a surface of the entrance slot **238**, the bottom circumferential surface **330** is configured to engage a surface of the circumferential locking slot **242**, and the bottom ramp surface **332** is configured to engage a surface of the ramp slot **240** of the center tube **206** as previously described herein.

In addition, the tab **322** further comprises a top ramp surface **334** extending from the first axial surface **328** that is parallel to the bottom ramp surface **332** for engaging another surface of the ramp slot **240**. A top circumferential surface **336** extends from the top ramp surface **334** that engages another surface of the circumferential locking slot **242**. A second axial surface **338** connects the bottom ramp surface **332** to the top circumferential surface **336**, and is configured to engage the stop surface **244** of the center tube **206**.

The pedestal **300** may be complementarily shaped to the bottom open end **222** of the center tube **206**. Consequently as seen in FIG. **5**, the pedestal **300** may have a bottom annular portion **318** that defines a bottom diameter **320**, and a top annular portion **312** that defines a top diameter **316** that is less than the bottom diameter **320**. Also, a flared annular

portion **324** (may also be referred to as a funnel annular portion) connects the top annular portion **312** to the bottom annular portion **318**.

The pedestal **300** may further comprise a canister portion **340** (e.g. may be attached thereto) and may include an annular outer wall **342** defining an outer diameter **344** that is greater than the bottom diameter **320** of the bottom annular portion **318** to form the annular cavity **118** (see FIG. 1). As best seen in FIG. 5, a support wall **346** that is configured to support the bottom annular portion **318**, and a funnel wall **348** connecting the annular outer wall **342** to the support wall **346**.

Looking at FIGS. 1, 7 and 8, it may be understood that the canister filter system **100** and its center tube **206** may be modified so that the center tube **206** has both top and bottom locking features for attaching both the base **102** and the canister **104** to the filter element **200**. As shown in FIG. 8, a top pedestal **300** may be provided that is identically configured to that of FIG. 5 except that it has been rotated 180 degrees about an axis passing through the axial midpoint of the center tube **206** and that is perpendicular to the longitudinal axis **216** of the center tube **206** so that the top pedestal **300** may be operatively associated with the base **102** instead of the canister **104** (e.g. attached to the base instead of the canister as previously described herein).

In FIG. 8, a bottom pedestal **300'** may be provided that includes an at least partially annular body **302'** defining a longitudinal axis **306'**, a radial direction **308'**, and a circumferential direction **310'** that are defined the same way as those of the top pedestal **300**.

The bottom pedestal **300'** may include a top annular portion **312'** that terminates at a top free end (not clearly shown but understood to be similar to what has been previously described for pedestal **300** or what is shown in FIG. 1), and that defines a top diameter (not clearly shown but understood to be similar to what has been previously described for pedestal **300** or what is shown in FIG. 1). Likewise, the bottom pedestal **300'** may also include a bottom annular portion **318'** defining a bottom diameter (not clearly shown but understood to be similar to what has been previously described for pedestal **300** or what is shown in FIG. 1) that is greater than the top diameter **316'**. A flared annular portion **324'** that connects the top annular portion **312'** to the bottom annular portion **318'**. Also, a bottom tab **322'** extends radially from the top annular portion **312'**.

The bottom annular portion **318'** may define at least one thru-slot (not clearly shown but understood to be similar to what has been previously described for pedestal **300**) that extends radially and possibly axially downwardly through the bottom annular portion **318'** similar to pedestal **300** as previously described earlier herein. It is to be understood that the pedestal **300** may be substituted for the bottom pedestal **300'** shown in FIG. 8 in some embodiments.

For example in FIGS. 1, 2 and 3, it is to be understood that the canister **104** may comprise an outer annular wall **124** that defines an outer diameter **126** that is greater than the bottom diameter of the bottom pedestal **300'** (shown in FIG. 8). A support wall **346** would support the bottom annular portion **318'** of the bottom pedestal **300'**, and an arcuate wall **130** would connect the support wall **346** to the outer annular wall **124**.

As best seen in FIGS. 2 and 3, the filter seal **120** would contact the arcuate wall **130**, forming a drain reservoir **132**. Moreover, the bottom open end **222** of the modified center tube **206** would impinge upon the filter seal **120**, creating a fluid tight seal between center tube **206** and the filter seal **120**, and another fluid tight seal between the filter seal **120**

and the arcuate wall **130**. The arcuate wall may have another shape such as conical, etc. The filter seal may be omitted in some embodiments such as when the canister has a bottom closed end, etc.

Referring back to FIG. 8, the center tube **206'** includes a bottom locking slot **224'** that extends axially from the bottom open end **222'** of the center tube **206'** an axial distance **226'**, and then extends circumferentially a circumferential distance **228'**. The center tube **206'** may also have a top locking slot **224''** that extends axially from the top open end **220'** of the center tube **206'** another axial distance **226''**, and then extends circumferentially in the opposite direction as compared to the bottom locking slot **224'**.

Furthermore, the center tube **206'** may define an inner diameter (not clearly shown but understood to be similar to what has been previously described for pedestal **300** or what is shown in FIG. 1) that is slightly greater than the top diameter of the bottom pedestal **300'** that is disposed within the inner diameter of the center tube **206'**. The bottom tab **322'** of the bottom pedestal **300'** is disposed in the bottom locking slot **224'** and is configured to guide the movement of the center tube **206'**. Also, the top pedestal **300** may define a diameter (e.g. **316**) that is slightly less than the inner diameter of the center tube **206'**, and that is disposed within the inner diameter of the center tube **206'**. The top tab **322** of the top pedestal **300** may be differently configured than the bottom tab **322'** of the bottom pedestal **300'**, and the top tab **322** may be disposed in the top locking slot **224''** to guide the movement of the base.

With continued reference to FIG. 8, a filter element with a center tube **206'** having both top and bottom locking features that may be used with the filter canister system just described will now be discussed. It is to be understood that the center tube **206'** shown in FIG. 8 is simplified to show the use of top and bottom locking features simultaneously. In actuality, the middle portion of the center tube **206'** would have perforations similar to what is shown in FIG. 7.

As alluded to earlier herein, the center tube **206'** may have a bottom open end **222'** that is joined to the center tube **206'** opposite the top open end **220'**, both of which are disposed along the longitudinal axis **306'**. The bottom open end **222'** may define a radially outer surface **232'**, and a radially inner surface that is in communication with the central reservoir (not clearly shown in FIG. 8).

A first locking feature **236'** may be disposed proximate to the bottom open end **222'**. The first locking feature **236'** may include a first entrance slot **238'** that is disposed on the radially inner surface (not clearly shown in FIG. 8) of the center tube **206'** extending axially upwardly from the bottom open end **222'**. A first ramp slot **240'** may extend axially upwardly and circumferentially counterclockwise from the first entrance slot **238'** along a first predetermined direction **241'**, the first ramp slot **240'** being in communication with the first entrance slot **238'**. The entry slot may be axially tapered to provide a lead-in during assembly.

Similarly, a second locking feature **236''** may be disposed proximate to the top open end **220'**. The second locking feature **236''** may include a second entrance slot **238''** that is disposed on the radially inner surface (not clearly shown in FIG. 8) of the center tube **206'** extending axially downwardly from the top open end **220'**. A second ramp slot **240''** may extend axially downwardly and circumferentially clockwise from the second entrance slot **238''** along a second predetermined direction **241''**, the second ramp slot **240''** being in communication with the second entrance slot **238''**.

As shown in FIG. 8, the first predetermined direction **241'** and the second predetermined direction **241''** are parallel to

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each other. For example, this may be true when the first locking feature **236'** is identically configured as the second locking feature **236"** when rotated about an axis that is perpendicular to the longitudinal axis **306'** an amount of 180 degrees, and then aligned axially and circumferentially with the second locking feature **236"**. This may not be the case for other embodiments of the present disclosure.

The first locking feature may further comprise a first circumferential locking slot **242'** extending circumferentially counterclockwise from the first ramp slot **240'**, the first circumferential locking slot **242'** being in communication with the first ramp slot **240'** and terminating at a first sloping surface **254** that extends axially upwardly and circumferentially counterclockwise.

Similarly, the second locking feature **236"** may further comprise a second circumferential locking slot **242"** extending circumferentially clockwise from the second ramp slot **240"**, the second circumferential locking slot **242"** being in communication with the second ramp slot **240"**, and terminating at a second sloping surface **254'** extending axially downwardly and circumferentially clockwise.

For the first locking feature **236'**, a first detent notch **246'** may extend axially upwardly from the first circumferential locking slot **242'** being in communication therewith. Likewise for the second locking feature **236"**, a second detent notch **246"** may extend axially downwardly from the second circumferential locking slot **242"** being in communication therewith.

The first detent notch **246'** may terminate circumferentially at a first stop surface **244'** that extends axially upwardly from the first sloping surface **254**, and the second detent notch **246"** may terminate circumferentially at a second stop surface **244"** that extends axially downwardly from the second sloping surface **254'**.

Looking at FIG. 8, the center tube **206'** has a necked down configuration at the bottom open end **222'** including an enlarged bottom annular portion **248'**, a reduced top annular portion **250'**, and a transitional annular portion **252'** therebetween. The entrance slot **238'** is formed by the enlarged bottom annular portion **248'**, the transitional annular portion **252'**, and the reduced top annular portion **250'**. The first ramp slot **240'** is formed by the reduced top annular portion **250'**. The first circumferential locking slot **242'**, and the first detent notch **246'** are formed by the reduced top annular portion **250'**. The top open end **220'** have been similarly described, being mirrored about an axial midplane of the center tube **206'**.

Referring to FIGS. 5 and 8, a pair of pedestals **300, 300'** that may be used with the filter element just described will be discussed in further detail.

Each of the pair of pedestals **300, 300'** may comprise an at least partially annular body **300, 300'** defining a longitudinal axis **306, 306'**, a radial direction **308, 308'**, and a circumferential direction **310, 310'**.

Each may have a top annular portion **312, 312'** terminating at a top free end **314**, and a tab **322, 322'** extending radially from the top annular portion **312, 312'**. The tab **322, 322'** may include a first axial surface **328, 328'**, a bottom circumferential surface **330, 330'**, and a bottom ramp surface **332, 332'** extending from the bottom circumferential surface **330, 330'**. As best seen in FIG. 8, the surface of one tab may be differently configured (e.g. have different dimensions) than the corresponding surface of the other tab but not necessarily so.

Still looking at FIGS. 5 and 8, each tab **322, 322'** may further comprise a top ramp surface **334, 334'** extending from the first axial surface **328, 328'** that is parallel to the

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bottom ramp surface **332, 332'**, a top circumferential surface **336, 336'** extending from the top ramp surface **334, 334'**, and a second axial surface **338, 338'** connecting the bottom ramp surface **332, 332'** to the top circumferential surface **336, 336'**.

Each pedestal **300, 300'** may further comprise a bottom annular portion **318, 318'** that defines a bottom diameter **320**, and the top annular portion **312, 312'** defines a top diameter **316** that is less than the bottom diameter **320**. A flared annular portion **324, 324'** may connect the top annular portion **312, 312'** to the bottom annular portion **320, 320'**.

All of these various surfaces of one tab may be differently configured than the corresponding surfaces of the other tab. Hence, the tab of each of the pair of pedestals may be differently configured.

As shown in FIG. 8, multiple locking features and tabs may be used at both ends of the center tube **206'**, but not necessarily so.

During installation for the embodiment in FIG. 8, the base and/or top pedestal is first attached to the center tube of the filter element. The top tab is prevented from entering the upper detent notch due to its dimensions. Then, the base and filter element are attached to the bottom pedestal until the bottom tab is seated in the lower detent notch since the bottom tab is smaller than the top tab.

Next, an interface for use with a filter system similar to that described earlier herein will be discussed with reference to FIGS. 9 thru 11.

The interface **400** may comprise a filter element **200** that is configured the same as or similar to that as previously described herein. The filter element **200** may include a first open end **402** defining a first locking slot **404**, and a second open end **406** defining a second locking slot **408**. A first pedestal **410** may be provided that includes a first tab **412** that is disposed in the first locking slot **404**. Similarly, a second pedestal **414** may be provided including a second tab **416** that is disposed in the second locking slot **408**. The second tab **416** may be differently configured as the first tab **412**. Either tab may have any suitable configuration including rectangular, square, elliptical, circular, quadrilateral, etc.

In some embodiments, the first locking slot **404** may be identically configured as the second locking slot **408** when rotating the geometry of the first slot about an axis that is perpendicular to the longitudinal axis **216** of the filter element **200** an amount of 180 degrees and then aligning the first locking slot **404** with the second locking slot **408** axially, circumferentially, and radially. This may not be the case in other embodiments of the present disclosure.

As illustrated in FIGS. 9 and 10, the first locking slot **404** may include a first detent notch **418** that defines an extremity **420** that is the furthest axially from the first open end **402** of the filter element **200**, and the first tab **412** may be configured to move axially into and out of the first detent notch **418**.

As best seen in FIG. 11, the second locking slot **408** may include a second detent notch **422** that defines an extremity **420'** that is furthest axially from the second open end **406** of the filter element **200**, and the second tab **416** may be configured to be prevented from moving into and out of the second detent notch **422**.

To that end, FIGS. 10 and 11 show that the first tab **412** includes a first bottom circumferential surface **424** defining a first bottom circumferential surface width **426**, and the second tab **416** includes a second bottom circumferential surface **428** defining a second bottom circumferential surface width **430** that is greater than the first bottom circumferential surface width **424**. The difference in these struc-

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tures along with the necked throat of the entrance of the detent notches, formed by an axial throat surface **432** and a slanted throat surface **434**, allows the first tab to enter the detent notch and not the second tab.

Also, the first tab **412** includes a first ramp surface **436** and a second ramp surface **438** that are parallel to each other and that define a first ramp width **440** that measured perpendicularly to the first ramp surface **436**. Likewise, the second tab **416** defines a third ramp surface **442** and a fourth ramp surface **444** that are parallel to each other and that define a second ramp width **446** that is measured perpendicularly to the third ramp surface **442** and the fourth ramp surface **444**. The second ramp width **446** may be greater than the first ramp width **440**, blocking its entry into the detent notch.

Accordingly, the first tab **412** may be configured to move axially (into the entrance portion of the first locking slot **404**), diagonally (along the ramp portion of the first locking slot **404**), circumferentially (in the circumferential portion of the first locking slot **404**) and axially into the first detent notch **418**, while the second tab **416** is configured to move axially, diagonally, and circumferentially in like manner as the first tab, except that the second tab **416** reaches a stop **448** (see FIG. 11) in the second locking slot **408** before entering the second detent notch **422**. This may not be the case for other embodiments of the present disclosure.

Any of the aforementioned features may be varied in configuration to be different in other embodiments of the present disclosure. In particular, the locking slot may follow any desirable path and may have differently configured walls that form them while the tabs may have any suitable configuration including round, polygonal, etc.

INDUSTRIAL APPLICABILITY

In practice, a filter element, a pedestal, or a canister filter system according to any embodiment disclosed herein may be obtained or provided in an OEM (original equipment manufacturer) or aftermarket context. The various features previously discussed may be used to both properly orient, position, and lock the various components of the canister filter system into place.

The center tube and the pedestal may be made from any suitable material including plastic, metal, etc. It may be desirable to choose materials that are chemically compatible with the fluids being filtered.

In previous designs, there may be a problem related to the attachment of the filter element to the housing (canister). Conventionally, the attachment of filter element to the housing, maintaining the proper force on the seal, and providing alignment of the filter element to the housing may be difficult.

Various embodiments of the present disclosure allow a newly developed method to install the filter element to the housing. The filter installation helps to ensure the proper alignment of filter element within the housing as well as maintaining the downward force on the seal. Further, the new design of filter element includes slots in the center tube that will allow tabs on the filter housing pedestal to engage. When the filter is rotated, it will provide locking of the filter element with the housing and the required force to help ensure the proper engagement of the seal with the housing base for the separation of pure and impure water, etc. Moreover, the slots and pedestal tabs can be changed (in number, position, configuration, etc.) so that wrong filter is not used resulting in damage to machine components.

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More specifically, the tab of the lower pedestal engages the corresponding locking slot of the center tube, providing downward force to push bottom seal into place when the filter is rotated into housing at installation. Also, the position of the top seal is also provided so that it will seal properly.

It is contemplated that other embodiments of the present disclosure may work or be structured differently so not all of the benefits just described may be obtained.

In some embodiments, the filter element/center tube may sit on top of the tab(s) at the bottom of the housing. In such a case, a proper seal may be provided at the top end, but the user may have to provide the force to overcome the seal at the bottom, etc.

It is further contemplated that the features of the pedestal including the tab may be swapped for the features of the center tube/filter element including the locking feature/slot, etc. in other embodiments of the present disclosure.

In light of the foregoing, a method of assembly of a canister filter system according to an embodiment of the present application may be employed as depicted in FIG. 12.

The method **500** may comprise inserting a first filter component into a second filter component (step **502**), and continuing the insertion until a first tab of either the first filter component or the second filter component contacts the other of the first filter component and the second filter component (step **504**). For example, the filter element may be inserted into a filter base or a canister (housing), etc. until the filter element contacts the tab of the filter base or canister (or vice versa).

In particular embodiments, the first tab of either the first filter component or the second filter component slides into a first slot of the other of the first filter component or the second filter component (step **506**).

Sometimes, the first tab slides upwardly in the first slot until the first tab reaches a ramp portion of the first slot (step **508**, e.g. see FIGS. 2 thru 4, 8).

Then, the first filter component or the second filter component may be rotated, causing the first tab to slide along the ramp portion of the first slot until the first tab reaches a circumferential portion of the first slot, causing the first filter component or the second filter component to contact a seal (step **510**, e.g. see FIGS. 2 thru 4).

Next, the first filter component or the second filter component (or both) may be rotated until the first tab reaches a stop or a detent notch (step **512**, e.g. see FIGS. 4, 10, and 12).

In some embodiments, the first tab slides upwardly or downwardly in the detent notch (step **514**, e.g. see FIGS. 4, 8, and 10).

In yet further embodiments, the method **500** may further comprising attaching the first filter component or the second filter component to a third filter component, providing a seal (step **516**, e.g. see FIG. 1 which indicates the canister and the filter element may be attached to the base to create one or more seals).

The method **500** may also include inserting a second tab of a third filter component into a second slot of the first filter component or the second filter component until the second tab reaches a ramp portion of the second slot (step **518**, e.g. see FIGS. 1, 8 thru 11).

In such a case, the method **500** may further comprising rotating the first filter component or the second filter component until the second tab slides along the ramp portion of the second slot, reaches a circumferential portion of the second slot, and continues until the second tab reaches a second stop or a second detent notch (step **520**).

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If a second detent notch is reached, then the second tab may slide upwardly or downwardly in the second detent notch (step 522).

Once assembled as best seen in FIG. 1 for some embodiments of the present disclosure, a bottom seal 140, an inner top seal 142, and an outer top seal 144 may be created to help prevent any fluid from leaking.

It will be appreciated that the foregoing description provides examples of the disclosed assembly and technique. However, it is contemplated that other implementations of the disclosure may differ in detail from the foregoing examples. All references to the disclosure or examples thereof are intended to reference the particular example being discussed at that point and are not intended to imply any limitation as to the scope of the disclosure more generally. All language of distinction and disparagement with respect to certain features is intended to indicate a lack of preference for those features, but not to exclude such from the scope of the disclosure entirely unless otherwise indicated.

Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein.

It will be apparent to those skilled in the art that various modifications and variations can be made to the embodiments of the apparatus and methods of assembly as discussed herein without departing from the scope or spirit of the invention(s). Other embodiments of this disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the various embodiments disclosed herein. For example, some of the equipment may be constructed and function differently than what has been described herein and certain steps of any method may be omitted, performed in an order that is different than what has been specifically mentioned or in some cases performed simultaneously or in sub-steps. Furthermore, variations or modifications to certain aspects or features of various embodiments may be made to create further embodiments and features and aspects of various embodiments may be added to or substituted for other features or aspects of other embodiments in order to provide still further embodiments.

Accordingly, this disclosure includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the disclosure unless otherwise indicated herein or otherwise clearly contradicted by context.

What is claimed is:

1. A filter element for a canister filter system comprising: an annular filter media defining a longitudinal axis; a tube arranged coaxially with the annular filter media; a first open end joined to the tube and having formed therein a locking slot; and a second open end joined to the tube opposite the first open end; the locking slot having including an entrance portion extending in an axial direction, a ramp portion extending diagonally from the entrance portion in the axial direction and in a circumferential direction, a circumferential portion extending in the circumferential direction, and a detent notch extending in the axial direction from the circumferential portion and forming an axial extremity of the locking slot;

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wherein the locking slot forms a necked throat between the circumferential portion and the detent notch; and wherein the locking slot includes an axial throat surface and a slanted throat surface.

2. The filter element of claim 1 wherein the slanted throat surface extends from the circumferential portion to the detent notch.

3. The filter element of claim 2 wherein a right angle corner is defined by the axial throat surface and an adjacent surface of the circumferential portion and is positioned opposite to the slanted throat surface.

4. The filter element of claim 2 wherein the axial extremity is defined by an axial extremity surface of the locking slot that extends in the circumferential direction.

5. The filter element of claim 1 wherein the second open end includes a second locking slot including an entrance portion extending in an opposite axial direction, a ramp portion extending diagonally from the respective entrance portion in the opposite axial direction and in an opposite circumferential direction, a circumferential portion extending in the opposite circumferential direction, and a detent notch extending in the opposite axial direction from the circumferential portion and forming an axial extremity of the second locking slot.

6. The filter element of claim 1 wherein the filter media forms a central passage and the tube includes a center tube arranged within the central passage.

7. A filter element for a canister filter system comprising: an annular filter media defining a longitudinal axis; a tube arranged coaxially with the annular filter media; a first end joined to the tube and having formed therein a locking slot; and a second end joined to the tube opposite the first end;

the locking slot having an entrance portion extending in an axial direction away from the first end and toward the second end, a ramp portion extending diagonally from the entrance portion in the axial direction away from the first end and toward the second end and in a circumferential direction, a circumferential portion extending in the circumferential direction, and a detent notch, the detent notch extending from the circumferential portion in the axial direction away from the first end and toward the second end, and forming an axial extremity of the locking slot spaced from the circumferential portion in the axial direction away from the first end and toward the second end;

the detent notch including an end edge forming the axial extremity and extending in the circumferential direction between a first axial edge and a second axial edge, and the first axial edge and the second axial edge each being oriented parallel to the longitudinal axis.

8. The filter element of claim 7 wherein the second end includes a second locking slot including an entrance portion extending in an opposite axial direction away from the second end, a ramp portion extending diagonally from the respective entrance portion in the opposite axial direction and in an opposite circumferential direction, a circumferential portion extending in the opposite circumferential direction, and a detent notch extending in the opposite axial direction from the circumferential portion and forming an axial extremity of the second locking slot.

9. The filter element of claim 8 wherein the axial extremity of the first locking slot and the axial extremity of the second locking slot are defined, respectively, by a first axial extremity surface that extends in the circumferential direction and a second axial extremity surface that extends in the opposite circumferential direction.

10. The filter element of claim 7 wherein the locking slot forms a necked throat between the circumferential portion and the detent notch, and includes an axial throat surface and a slanted throat surface.

11. The filter element of claim 7 wherein at least one of the first end or the second end includes an open end.

12. A filter element for a canister filter system comprising:

an annular filter media defining a longitudinal axis;

a tube arranged coaxially with the annular filter media;

a first end joined to the tube and having formed therein a locking slot; and

a second end joined to the tube opposite the first end;

the locking slot having an entrance portion extending in

an axial direction away from the first end and toward

the second end, a ramp portion extending diagonally

from the entrance portion in the axial direction away

from the first end and toward the second end and in a

circumferential direction, a circumferential portion

extending in the circumferential direction, and a detent

notch, the detent notch extending from the circumferential

portion in the axial direction away from the first

end and toward the second end, and forming an axial

extremity of the locking slot spaced from the circumferential

portion in the axial direction away from the

first end and toward the second end;

wherein the locking slot forms a necked throat between

the circumferential portion and the detent notch, and

includes an axial throat surface and a slanted throat

surface.

* * * * *